

Session 1: AI for Coding – Tools, Concepts, and Setup

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What We Will Cover (Sessions 1 & 2)

- ▶ Part 1: Tools, concepts, and setup
 - Overview of AI coding tools and how to choose among them
 - How to use AI Agents: Terminal, Desktop App, VS Code
 - Conceptual overview of context engineering with Claude
 - VS Code and Claude Code integration
 - Git and Github: why they matter for research and Claude

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 - Conceptual overview of context engineering with Claude
 - VS Code and Claude Code integration
 - Git and Github: why they matter for research and Claude
- ▶ Part 2: Practical workflows
 - Writing, debugging, and automating code with AI
 - “Vibe coding” from plain English descriptions
 - Project organization and replication packages
 - Demo: data cleaning script from scratch (python and stata)

Coding Perhaps Highest-Value Use of AI

- ▶ AI most reliable when output is verifiable: code runs or it does not
- ▶ Economists spend enormous time on repetitive coding tasks
 - AI can handle most of them
- ▶ Massive productivity gains across skill levels:
 - Beginners: “vibe code” to write code in an unfamiliar language (no syntax!)
 - Advanced users: automate drudge work, prototype faster, reduce bugs

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 - Advanced users: automate drudge work, prototype faster, reduce bugs
- ⇒ First two sessions will cover how to use AI to complete coding tasks

Overview of AI Coding Tools

The AI Coding Landscape: Many Options

- ▶ General-purpose AI assistants (good at coding, among many other things):
 - **Claude** (Anthropic) — our focus today
 - **ChatGPT / GPT-4o** (OpenAI)
 - **Gemini** (Google)
- ▶ Agentic coding tools (read files, write and run code autonomously):
 - **Claude Code** (Anthropic) — our focus today
 - **Codex** (OpenAI) — closest competitor to Claude Code
 - **GitHub Copilot** (Microsoft) — inline suggestions in editor
 - **Cursor / Windsurf** — AI-first code editors
- ▶ How to choose: task type, privacy needs, and workflow integration

Chatbot vs. Agentic Tool

- ▶ **Chatbot** (Claude.ai, ChatGPT): you ask → it answers → *you* act
- ▶ **Agentic tool** (Claude Code): you give a goal → it reads files, writes code, runs it, fixes errors, iterates → reports back
- ▶ For economists: instead of “here is some code,” it runs the code and tells you what it found

Which One Should You Use? Claude Code vs. Codex

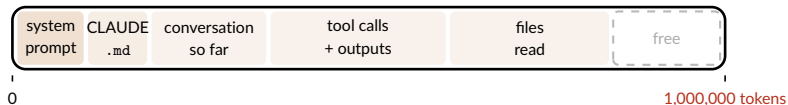
- ▶ Two serious options for agentic coding today: **Claude Code** and **Codex**
- ▶ Plans mirror each other at each price point:
 - \$20/mo: Claude Pro vs. ChatGPT Plus
 - \$100/mo: Claude Max (5x) vs. ChatGPT Pro (5x)
 - \$200/mo: Claude Max (20x) vs. ChatGPT Pro (20x)
- ▶ Both use a 5-hour rolling window + weekly cap; limits reset automatically
- ▶ At the same price, Codex currently gives more usage per dollar

Which One Should You Use? Claude Code vs. Codex

- ▶ Two serious options for agentic coding today: **Claude Code** and **Codex**
- ▶ **Codex** has some advantages:
 - More usage per subscription dollar — important for heavy users
 - Sometimes stronger on pure coding benchmarks
- ▶ **Claude Code** has some advantages:
 - Often better at tasks beyond code: understanding context, writing, reasoning
- ▶ **Bottom line:** this comparison is fast-moving and will keep changing
 - I am using Claude Code and find it excellent for my workflow
 - I would switch only if Codex proved clearly and consistently superior
 - My advice: pick one, learn it well, reassess in 6 months

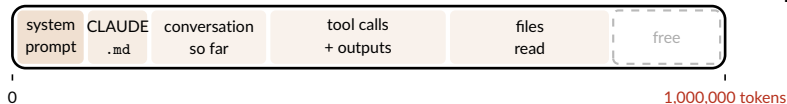
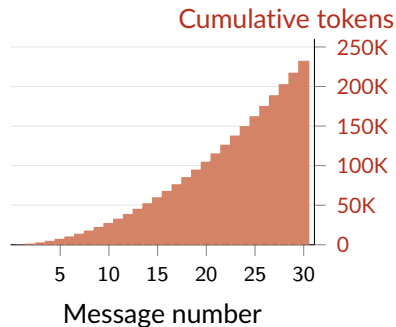
What You Pay For: Tokens and Context

- ▶ **Token:** the unit AI reads and writes — roughly $\frac{3}{4}$ of a word
 - **You pay for:** tokens *in* (prompt + files) and tokens *out* (reply) — input is cheaper
- ▶ **Context window:** everything the model holds at once — fills as session grows



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Getting the Most Out of Your Subscription

- ▶ Subscriptions have rolling limits — fewer tokens per task means more use
- ▶ How to use fewer tokens:
 - Start a new session for each new task: context accumulates and raises costs
 - Store standing instructions once so you never repeat yourself
 - Be specific: vague prompts waste tokens and invite unfocused replies
 - Ask multiple things at once so less back and forth
 - Compact your chats/context window
- ▶ Silly trick: start Claude when wake up so 5-hour window ends earlier in day

Contex Engineering with Claude Overview

Context Engineering

▶ Context Window

- ▶ Each prompt to Claude Code bundles the entire history of your session
 - ▶ your prompts, Claude's responses, files it read, code outputs, system prompts
- ▶ These are all appended until the context window hits its fixed size limit
 - ▶ For Claude Code this is around 200k tokens
- ▶ Claude then “**auto-compacts**” writes its own summary and then restarts fresh
 - ▶ less preferable to frequent intentional compaction

▶ Context Engineering

- ▶ You can tell Claude when to compact using `/compact your instructions`
- ▶ Develop habits that the context window focussed on relevant information
 - ▶ Claude.md vs skills —and— ‘ask before edits’ vs ‘plan’ Modes

Read more at [Paul GP's Guide](#)

Anatomy of a context window (example from Paul GP's Guide)

```
<|start|>system<|message|>You are ChatGPT, a large language model trained by OpenAI.  
Knowledge cutoff: 2024-06  
Current date: 2025-06-28  
Reasoning: high  
# Valid channels: analysis, commentary, final.  
Channel must be included for every message.  
Calls to these tools must go to the commentary channel: 'functions'.<|end|>
```

system prompt
(written by openAI)

ANATOMY OF A
MODERN LLM
CHAT

```
<|start|>developer<|message|># Instructions  
Be concise and technical. Provide code examples when relevant.
```

developer
message
(tool definitions)

```
# Tools  
## functions  
namespace functions {  
  type web_search = (_: {  
    query: string,  
    num_results?: number,  
  }) => any;  
} // namespace functions<|end|>
```

user
prompt

```
<|start|>user<|message|>What's new in the latest version of polars?<|end|>
```

thinking

```
<|start|>assistant<|channel|>analysis<|message|>User asking about latest polars version.  
My knowledge cutoff is 2024-06, need to search for recent updates.<|end|>
```

```
<|start|>assistant<|channel|>commentary<|message|><|call|>functions.web_search  
{  
  "query": "polars python latest version new features", "  
  "num_results": 5  
}<|end|>
```

tool call

```
<|start|>tool<|message|>[{"title": "Polars 1.8.0 Release Notes", "snippet": "New  
streaming engine, improved join performance, added pivot_wider...", "url": "..."}]<|end|>
```

tool
results

```
<|start|>assistant<|channel|>final<|message|>Polars 1.8.0 (latest as of Nov 2025)  
includes:
```

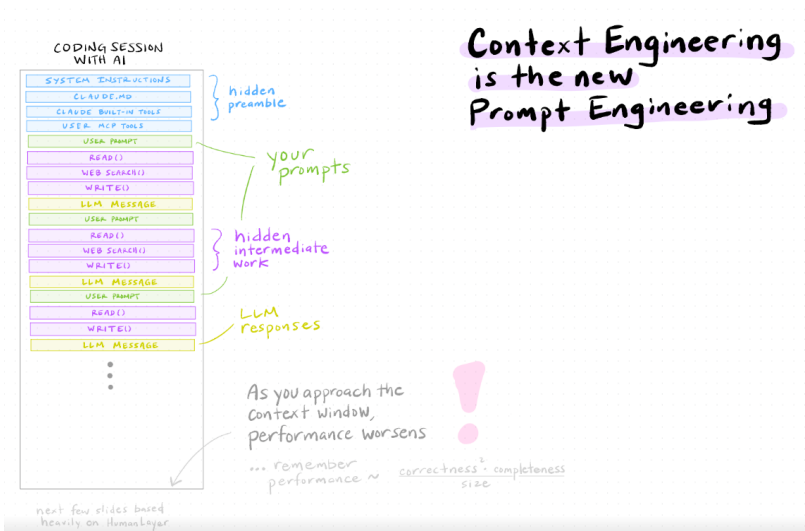
- **Streaming engine overhaul**: Better memory efficiency for large datasets
- **Join performance**: 2-3x faster for certain join types
- **New API**: `pivot_wider()` for reshaping data
- **LazyFrame improvements**: Enhanced query optimization

final response

```
Check the release notes for migration details if upgrading from older versions.<|return|>
```

You
typically
see only
these
two!

Context engineering (example from Paul GP's Guide)



Context auto-compact (example from Paul GP's Guide)

If you hit the
content window
the LLM will
auto-compact



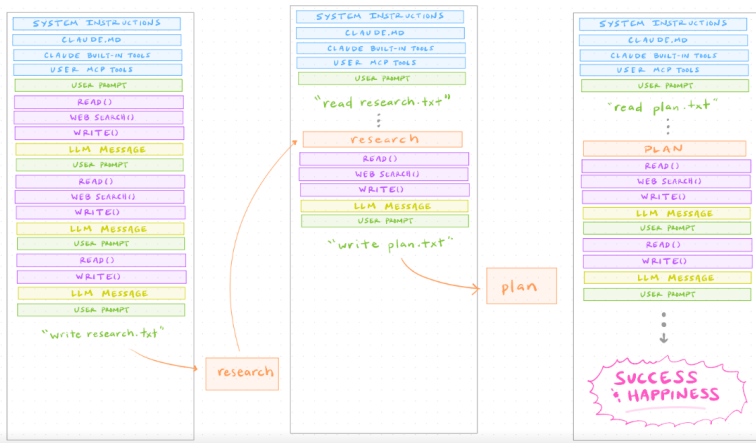
the model summarizes your conversation

...this is
rarely
ideal...



Context intentional compacting (example from Paul GP's Guide)

Frequent intentional compaction is better



CLAUDE.md and Skills

- ▶ **CLAUDE.md** (memory) — Claude reads at the start of every session
 - Store context that is always true so you don't repeat yourself in every prompt
 - `.claude/CLAUDE.md` is global, `your-project/CLAUDE.md` is *project* specific
 - Ask Claude to update after substantial changes to project
- ▶ **Skills** — Claude reads every time it is called
 - Store context for a task that you repeat often, but not always relevant
 - `.claude/skills/your-skill/SKILL.md` is called with `/your-skill`
 - Write and update your own skills or download from others

Read more at [Claes Backman's Guide](#) (full descriptions, examples of CLAUDE.md and skills, download pre-made skills)

your-project/CLAUDE.md (example from Claes Backman's Guide)

```
# Project: [Paper title]
```

Data

- Unit of analysis: firm-year panel, 2010-2022
- Raw data lives in data/raw/ – never edit these files
- Main dataset after cleaning: data/clean/panel.dta (N $\approx$ 47,000)

Code conventions

- R scripts are numbered (01_clean.R, 02_analysis.R, ...)
- All regressions cluster SEs at the firm level
- Use fixest for all panel regressions

LaTeX

- Main file: paper/main.tex
- Tables generated by 03_tables.R go to paper/tables/
- Use `\widehat{}` not `\hat{}` for estimators

`.claude/skills/robustness/SKILL.md` (example from Claes Backman's Guide)

```
---  
name: robustness  
description: Run standard robustness checks on the main regression.  
---
```

For the main specification in this project:

1. Re-run with alternative clustering (industry-year instead of firm)
2. Re-run dropping the top and bottom 1% of the outcome variable
3. Re-run on the pre-2020 subsample only
4. Produce a summary table comparing coefficients across all specs

Claude Modes

▶ Ask before edits

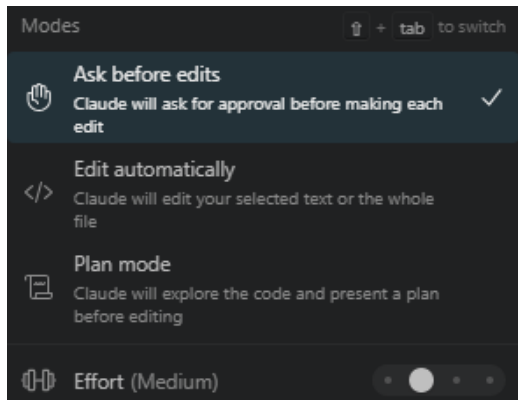
- ▶ Do steps one at a time
- ▶ Asks your permission before running each and every command

▶ Edit automatically

- ▶ Same as above but runs commands without asking permission (risky)

▶ Plan

- ▶ Claude writes plan.md with many steps
- ▶ You review and execute entire plan



Other tips for managing the context window

- ▶ **Use @-mentions to be selective**

- ▶ Rather than Claude reading everything, point to a specific file
- ▶ e.g. `update all the pathnames in @Lecture_2.tex`

- ▶ **Check usage with /context**

- ▶ Type `/context` in the panel to see a breakdown of how your context window is being used and how much space remains.

Files and Data that Claude is good at reading

- ▶ Claude Code works best with plain text files
 - ▶ e.g. `.txt`, `.do`, `.py`, `.tex`, `.csv`, `.log`, `.md`, etc.
- ▶ Tips for optimal performance with other file types
 - ▶ `.docx`: convert to markdown with `pandoc file.docx -o file.md`
 - ▶ `.xlsx`, `.dta`, `.rds`, `.sav`: convert to CSV first with a short script
 - ▶ `.pdf`:
 - ▶ Use the `.tex` file whenever possible (especially if equations and tables)
 - ▶ Long PDFs (50+ pages): convert to markdown first
 - ▶ Scanned PDFs: run OCR first, then convert to markdown
- ▶ Create a skill that loads file types, e.g. `/pdf-to-markdown`

Read more at [Claes Backman's Guide](#) (details on handling different file types)

Claude Code / Codex Interfaces

Three Ways to Use an Agentic Coding Tool

▶ **Terminal** (command line):

- Most powerful; works directly in your project directory
- Best for: running scripts, data analysis, replication tasks

▶ **Desktop App:**

- Friendlier interface; easier to get started
- Projects feature: persistent context across sessions
- Best for: drafting, quick questions, early exploration

▶ **VS Code integration:**

- Agent lives inside your editor — see code and AI side by side
- Best for: sustained development work, Git integration, Jupyter/Python

Terminal vs. VS Code: Quick Comparison

Terminal (CLI)

- ▶ Lightweight; fast startup
- ▶ Works anywhere, incl. remote servers
- ▶ Keyboard-driven; fast for quick tasks
- ▶ No inline diffs; steeper setup

VS Code Extension

- ▶ Inline diffs; accept changes block-by-block
- ▶ Code & AI panel side by side
- ▶ Best for large codebases, sustained work
- ▶ Heavier on CPU and memory

Live Demo: Terminal

Live Demo: Desktop App

Live Demo: VS Code

VS Code with Claude Overview

What is VS Code?

- ▶ Integrated Development Environment (IDE)
 - ▶ Free, open source (created by Microsoft), and very popular
- ▶ Core features:
 - **Explorer:** file browser and project tree
 - **Editor:** syntax highlighting for every language (Stata, R, Python, $\text{\LaTeX}$, ...)
 - **Integrated terminal:** run scripts without leaving the editor
 - **Extensions:** Git, Claude, Stata, latex, python, etc. [Starter Pack for Economists](#).
 - **Command Palette** (`Ctrl+Shift+P`): search and run any action by name
- ▶ Claes Backman's Guide [Claude Code in VS Code for Academic Economists](#)
 - ▶ Other resources: [Official intro videos \(Microsoft\)](#), [Tips and Tricks \(Microsoft\)](#)

GWU_AI_study_group > slides > [Lecture_1.pdf](#)

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 - **Integrated terminal:** run scripts without leaving the editor
 - **Extensions:** Git, Jupyter, Claude Code, and more
 - **Command Palette** (`Ctrl+Shift+P`): search and run any action by name
- ▶ **Resources:** [Official intro videos \(Microsoft\)](#), [Tips and Tricks \(Microsoft\)](#)

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VS Code with Claude Overview

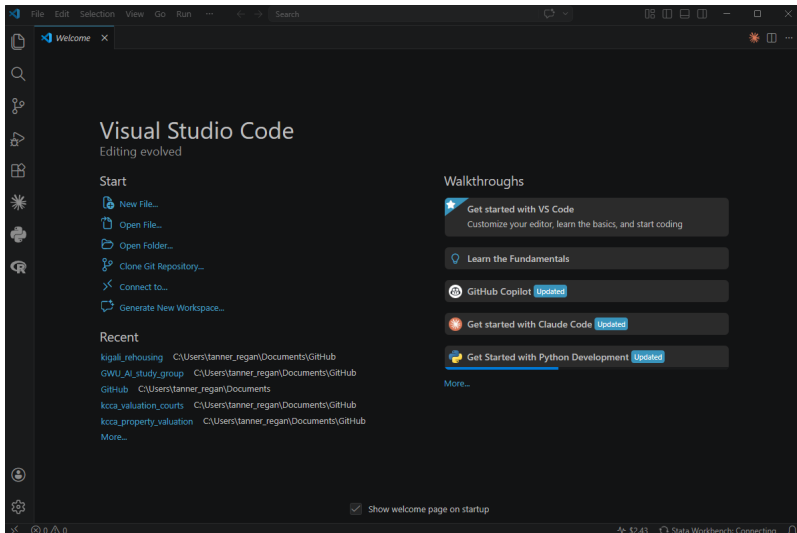
- The slide now uses Beamer's `columns` environment — list on the left half, image on the right half, both taking 48% of the text width with a small gap between them.

 **Ask before edits**

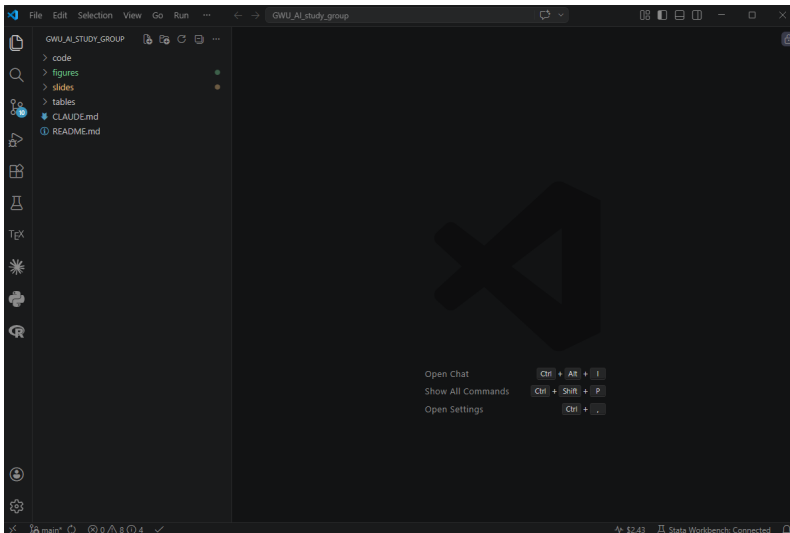
Claude Code **extension** in VS Code

- ▶ Give Claude access to your whole project directory to maximize its usefulness
 - ▶ Open your full project folder in VS Code (File → Open Folder)
 - ▶ This can be any folder on your computer — plain folder, a git repository, a Dropbox folder, etc.

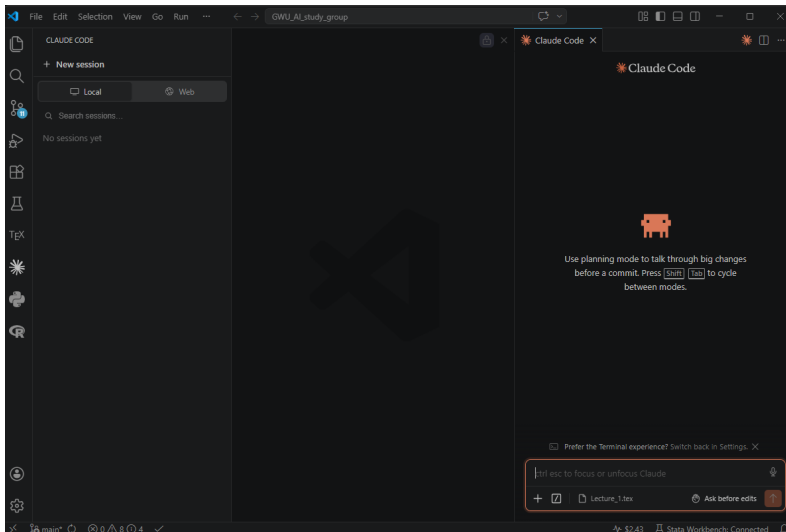
New VS Code Window — open new folder



Project folder is open — now open a new Claude Code session

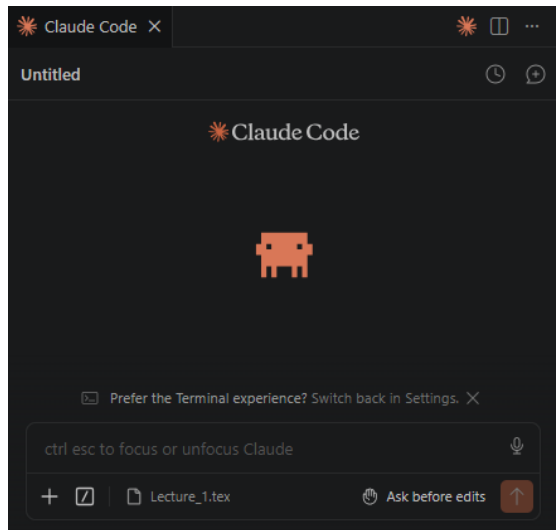


New Claude Code session is open



Claude Code extension in VS Code

- ▶ Each new conversation in the Claude panel is a fresh context
 - ▶ Click the + button at the top of the Claude panel to start one,
 - ▶ or type `/clear` to reset
- ▶ Go over a fresh window →
 1. Session Title
 2. Session History
 3. Mode menu
 4. Current File Selection
 5. Attachments
 6. Command Menu



Git: Version Control for Coding Projects

Why Git Matters for Research

- ▶ Version control = a complete history of every change to your code
 - ▶ No more: `analysis_v3_FINAL_revised2.do`
- ▶ Enables:
 - Backup: retrieve any prior version of your code
 - Collaborate: coauthors and RAs edit without overwriting each other
 - Track: what changed and who made the changes at every stage
- ▶ Can help improve project replicability both internally and externally
 - ▶ an automatic audit trail
 - ▶ sharing and iterating on projects publicly

Read more at Luís Fonseca's [Version Control with git \(for economists\)](#)

Git and Github Concepts

- ▶ Key concepts to remember
 - ▶ **Repository (repo):** a folder whose history Git is tracking
 - ▶ **Commit:** a snapshot of the code at a point in time, with a descriptive message
 - ▶ **Sync:** Pushes and pulls at the same time, so both repos are up to date
- ▶ Slightly more advanced concepts (learn these when you need them)
 - ▶ **Push:** sends changes up from your local repo to online/remote repo
 - ▶ **Pull:** brings changes down to your local repo from online/remote repo
 - ▶ **Branch:** a parallel version of the code (useful for exploration and RA work)
 - ▶ **Merge:** fold a branch back into the main code
 - ▶ **Pull request:** propose to merge one branch to another (e.g. main)
 - ▶ **Clone:** copy an online repo to local computer for initial setup
 - ▶ **Fork:** copy an online repo to your own GitHub account

Git vs. GitHub: Two Separate Things



git — the software that runs locally

- ▶ All history lives in the `.git` folder
- ▶ Records *commits*, *branches*, *merges*
- ▶ Compares any two versions line-by-line; flags conflicts



GitHub — online hosting platform

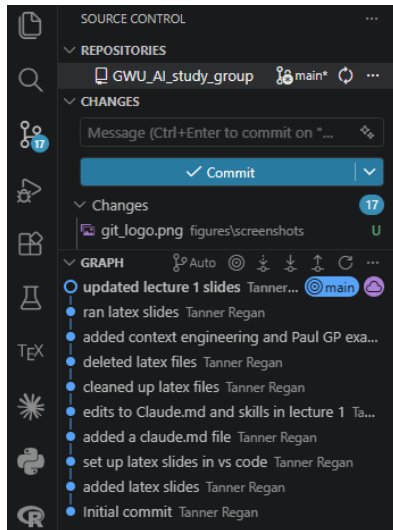
- ▶ Stores a remote copy of your repo
- ▶ Alternatives: GitLab, Bitbucket
- ▶ Remote endpoint for *Push*, *Pull*, *Clone*; introduces *Fork*

Typical Git workflows for an economist

- ▶ Typical workflow for an economist (alone or with trusted RA)
 1. Pull/Sync code at start of work session (get latest version of online repo)
 2. Work on code; when something works, *commit* it with a descriptive message
 3. Push/Sync at end of work session (updates the online repo)
- ▶ More elaborate, with explicit RA review
 1. RA syncs and then creates a new branch
 2. RA works on branch, makes commits and makes a pull request
 3. You review the pull request (compare the branch to main)
 - ▶ recall git highlights differences line by line and flags conflicts
 4. You merge back into *remote* main branch, or ask RA for fixes
 5. Pull/Sync to update your local repo

Git source control panel in VS Code

- ▶ Source Control panel
 - ▶ Click the github icon to open
- ▶ Go over an example →
 1. Icon with number of changes
 2. Message bar for commit
 3. Pull/Sync button
 4. commit button (also push)
 5. List of changes
 6. Graph of commits and branches



Setting up a github repository

- ▶ Go to github.com
 - ▶ Create an account, add new repository, add collaborators
 - ▶ Create a readme file that describes the project
- ▶ Open VS Code git extension (and sign into your GitHub account)
 - ▶ clone the repo to local computer: `Ctrl + Shift + P` (Windows/Linux) or `Cmd + Shift + P` (Mac) to open the Command Palette. Type `Git: Clone` and press Enter. Choose from the list of your repos.
- ▶ I have created a GitHub repo for our workshop [GWU_AI_study_group](#)
 - ▶ It is setup as Public, so you can already look at it and even Fork it
 - ▶ If you make a GitHub account and send me your username, I will add you as a collaborator so you can push changes to the repo as well

What Goes in a Repository?

- ▶ **Do NOT** store any datasets (source or generated)
 - instead store in **Box** (GW-endorsed secure storage)
 - or locally, Dropbox, Google Drive, etc.
alternatively, write a `.gitignore` file that tells git to ignore specific subfolders (e.g. `/data`)
- ▶ **Do** store all other project materials
 - code (Stata, Python, $\text{\LaTeX}$)
 - output: tables, figures
 - CLAUDE.md, notes, slides

```
GWU_AI_study_group/  
├── slides/  
│   ├── _preamble.tex  
│   ├── Lecture_1.tex  
│   └── Lecture_2.tex  
├── code/  
├── figures/  
├── notes/  
├── tables/  
├── CLAUDE.md  
└── README.md
```

Git and Claude Code Integration

- ▶ Claude Code is get-aware
 - ▶ When the open project is a git repo, Claude can see the entire commit history
- ▶ Claude can facilitate git operations
 - ▶ Write commit messages `commit my changes with a descriptive message`
 - ▶ Explain what changed `what changed between commit A and B?`
 - ▶ Resolve a merge conflict `fix RA code to merge with the main branch`

Read more at [Claes Backman's Guide](#)

Setting Up Your Environment

Core tools needed to set yourself up

- ▶ Software to install (first two are substitutable)
 1. **VS Code** — free to download [here](#)
 - ▶ + Claude Code extension
 2. **Claude Code** — free to download [here](#)
 3. **Git** — free to download [here](#)
- ▶ Subscriptions and accounts to create
 1. **Claude Code** — subscribe for a fee ([here](#))
 2. **GitHub** — free account ([here](#)), and send Tanner your username!
- ▶ Guides
 - ▶ A step-by-step setup guide for Mac ([here](#)) and Windows ([here](#))
 - ▶ However, Claude can walk you through any setup as long as you:
 - ▶ install VS Code + Claude Code extension
 - ▶ or install Claude Code desktop app

Other set up if you want to follow along next class

- ▶ Software to install (first two are substitutable)
 1. **Miniconda** — free to download [here](#), python will come included
 2. **MikTeX** — free to download [here](#), I also had to install [Strawberry Perl](#)
 3. **Stata** — our old friend
- ▶ VS Code Extensions
 - ▶ Claude Code, LaTeX Workshop, Stata Workbench, Python, Jupyter
 - ▶ See also Claes Backman's [Starter Pack for Economists](#)

Session 1 — Key Takeaways

1. AI has perhaps most value added in coding — output is verifiable
2. **Agentic** $\neq$ **chatbot**: reads files, writes and runs code, fixes errors, reports back
3. Two serious options today: **Claude Code** and **Codex** — pick one
4. **Manage your context window**: tokens accumulate fast — start fresh sessions, compact often, use CLAUDE.md so you never repeat yourself
5. **Git + GitHub** are your safety net — commit before big changes, push often
6. Three interfaces (Terminal, Desktop, VS Code) — VS Code integrates everything

Questions?

- ▶ Questions on tools, concepts, or setup?
- ▶ Thoughts on how this fits your current workflow?

Clarifications from day 1:

- ▶ Context window $\neq$ agent.
 - ▶ Can ask the 'main' agent, to spawn **subagents** to answer a prompt. Distinct from **agent teams** where multiple claude sessions work together
- ▶ **Cowork vs Claude Code**
 - ▶ Cowork comes with the desktop app
 - ▶ It restricts which IP addresses the agent can reach (can adjust permissions)
 - ▶ It's less risky, but less powerful than CC due to the restrictions

Questions from workshop 1, consider exploring in later sessions:

1. How can we get 'stable' or 'replicable' results from Claude or LLMs in general?
 - ▶ Partly why CC is attractive - code is verifiable and generates replicable results
 - ▶ What is the state of the literature on this?
 - ▶ Can 'temperature' be set with Claude Code?
2. Compare Codex and Claude more closely (Paul GP uses both simultaneously)
3. Accessing a remote server and using Claude in same workflow.
 - ▶ Does Claude need to be installed on the server? I think no.
 - ▶ How to set up VS Code with a remote server
 - ▶ both Claes Backman and Paul GP mention this
 - ▶ Working with GW resources: [CAAREN](#), [High Performance Computing \(HPC\)](#), [Research Cloud](#), [Research Storage \(NAS\)](#), [Virtual Computer Lab \(VCL\)](#) , e.g. [Stata](#)